Adaptive Cruise Controller for RC Car Using STM32

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***Abstract*—This project documents the design and developement of an RC Car is built from the ground up, as an attempt to implement and test the Adaptive Cruise Control System. Two STM32 Nucleo-Boards from STMicroelectronics are used for the development. The STM32L432KC is used for the transmitter and the STM32H723ZG for the RC Car system. A Python User Interface is developed for controllering the RC Car also perform data acquisition in order to analyze and develop the adaptive cruise controller.**

The sensor integrated to the RC Car is LIS3DH accelerometer, and Garmin LIDAR Lite V3 to measure the physical data necessary from the RC Car to develop the Adaptive Cruise Controller. The accelerometer was used to determine the velocity of the RC Car, and the LIDAR as an attempt to measure relative distance and velocity of another RC Car ahead. During the test there were challenges to develop the Adaptive Cruise Controller: the velocity calculated by the accelerometer started to drift from the true velocity as time increased due small noise, and the NRF24L01 trasnmit every 20ms was losing some communication for 1-1.5s which affect velocity calculation and started to drift.

Due to Earth's gravitional influence on the accelerometer, a calibration was attempted but the communication cut still affected to determine if the calibration was effective. Based on the test, the adaptive cruise controller could not be implemented, and implementing more sensors could provide more parameters to accurately calculate the velocity of the RC Car. In the future work, implement more sensor such as the Hall Effect senosr on the wheels and applying the Kalman Filter for sensor can improve velocity accuracy to enable closed-loop speed control.

***Index Terms*—Adaptive Cruise Control, STM32, PID Controller, LIDAR, NRF24L01, PyQt5, UART.**

I. Introduction

The purpose of this project is to explore electronics, software development, sensor integration, numerical analysis, real time data acquisition and primarily control engineering by building an RC Car from the ground up. An RC Car is designed to develop an Adaptive Cruise Controller with data acquisition in order to properly implement it. In Figure 1 shows a high-level overview of the entire project, the RC Car will listen and send data back to the transmitter in which is connected to a Python User Interface. The RC Car and transmitter are communicating with each other using the NRF24L01 Programmable RF module and the transmitter is connected to the computer by USB using the UART protocol and will either send commands to the transmitter or receive data. The setup of the Transmitter and Python User Interface is shown in Figure 2 whereas the RC Car and transmitter in Figure 3.

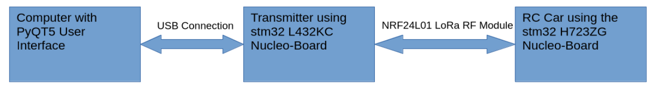

Fig. 1. High-level overview of the project architecture.

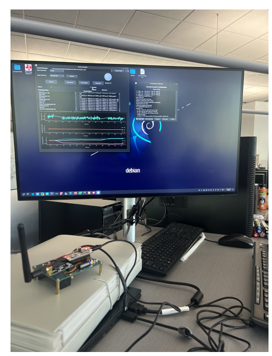

Fig. 2. The Transmitter connected to the python user interface.

Fig. 3. The Transmitter and RC Car.

II. Hardware Architecture

Your hardware section goes here.

A. RC Car Node

Subsection content.

B. *Transmitter Node*

Subsection content.

III. Firmware Design

Your firmware section goes here.

IV. Python Graphical User Interface

Your GUI section goes here.

V. Experimental Results

Your results go here.

VI. Conclusion

Your conclusion goes here.

References

[1] STMicroelectronics, *STM32H723ZG Datasheet*, Rev. 4, 2022.

[2] Garmin Ltd., *LIDAR-Lite v3 Operation Manual*, 2017.